

Jonathan Frias

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Petaluma, CA (707) 779-2019

EDUCATION

California State University, East Bay

Expected Graduation: May 2026

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures, Algorithms, Statistics, Linear Algebra, Computer Architecture, Programming Concepts, Software Responsibility, Computer Networks

SKILLS

Languages: TypeScript, React, C++, Python, JavaScript, Assembly, Rust, HTML/CSS

EXPERIENCE

Automation & Software Development Engineer Intern

LAWRENCE BERKLEY NATIONAL LAB

June 2025 - August 2025

- Develop a web interface to streamline organization of bar samples and configurations for X-ray spectroscopy scans, reducing manual data entry errors and saving time during experiments.
- Integrate with SciCat's backend servers to enable user authentication and dynamic retrieval of sample data, which is then parsed into structured scan files for LabView API integration.
- Automate scan file generation via Python, allowing users to pre-load experiment parameters and execute scans with a single click, eliminating the need for manual file creation during beamline operations.
- Enhance efficiency by enabling batch uploads of sample configurations, reducing experiment setup time and minimizing human error in high-stakes research environments.

Undergraduate Research Assistant

January 2025 - October 2025

CAHSI | WALKAWARE

Hayward, CA

- Develop an Android app using computer vision to detect sidewalk damage and provide spoken directional guidance for visually impaired users.
- Conduct data curation preprocessing to ensure high-quality datasets for machine learning models.
- Leverage Colab, Python, and a retrained version of OpenAI's GPT-4o Mini to process images, classify walkway hazards, and generate natural language alerts.
- Work on LLM-based narrative generation to enhance mobility through spoken instructions.

Field Service Engineer

August 2024 - May 2025

MURRAY WINDOW & DOOR

Campbell, CA

- Developing a maintenance schedule, contributed to 15% reduction in product failures.
- Participate in equipment installation, trial runs, investigative tests, repairs and overhauls.
- Maintain and update logs and databases to document equipment performance metrics, ensuring accurate records for analysis and reporting.
- Collaborate with cross-functional teams to resolve technical challenges and service processes.

PROJECTS

• **Fantasy Combat Game** | *Personal Project*

Utilized C++ standard library templates, OOP and custom header and implementation files. Program uses a random number generator to deal attack damage. Program will display the users health percentage along with declaring the winner.

• **Amazon Dupe Website** | *Personal Project*

Allows users to select items from the website and add them to their cart with their name, number of items, pricing, shipping cost, tax, and total expenses (utilize HTML, CSS, and JavaScript).

- **To-Do list** | *Personal Project*

Developed a web app to-do list application using Rust for backend functionality and JavaScript for the frontend, enabling efficient task management.

LEADERSHIP EXPERIENCE

- **MESA-U 2024 Hackathon** | *Best Research* | 100 CSU East Bay students

Implemented self automated message using Google Maps & WhatsApp API. Used to automatically send user information, medical allergies, current location to emergency contacts through WhatsApp. Being further developed to become a motorcycle safety alert system.

- **CAHSI** (Computing Alliance of Hispanic Serving Institution)

REU Program conducting research in computer vision under Lynne Grewe at CSU East Bay.